4D Historical Building Database Specification

COMPREHENSIVE DATA COLLECTION & TECHNICAL SCHEMA GUIDELINES FOR DOWNTOWN CHICAGO 3D RECONSTRUCTION

This document specifies the complete operational and technical schema for researchers, spatial data engineers, and architectural archivists tasked with compiling the **Historic Building Information Model (HBIM)** for downtown Chicago. The system supports a 4D spatial engine (3D geometry + continuous temporal scrub) allowing users to interactively navigate downtown Chicago at any point in history.

1. Architectural Strategy & Temporal Modeling Framework

In a 4D environment, buildings cannot be modeled as static database rows. Properties undergo continuous transformations across centuries: vertical expansions, architectural recladding, interior adaptive reuse, lot mergers, total destruction (e.g., the 1871 Great Fire), and redevelopment.

THE "EFFECTIVE-DATED STATE" ARCHITECTURE (SLOWLY CHANGING DIMENSIONS)

The database employs a dual temporal mechanism:

- **Effective Date Ranges (`valid_from` / `valid_to`):** Every mutable property attribute (height, façade material, tenant, name, architectural style) is anchored to a discrete version lifespan.
- **Lot Succession Linkages (`predecessor_id` / `successor_id`):** Structures occupying the same physical plot across time are explicitly linked, enabling clean rendering swaps when moving through time.
- **Runtime Snapshot Engine:** The 3D view executes queries using a `get_snapshot(target_year)` function returning all active entity states where `valid_from <= target_year < valid_to`.

2. Primary Schema Specifications — Core Entity (`Property`)

A. Identification & Location (Immutable Plot Attributes)

Fixed geographic attributes tied to the plot of land regardless of structural changes above.

Attribute Name	Data Type	Specification & Data Collection Rules
`property_id`	`UUID` (PK)	Immutable global primary key representing the physical plot of land.
`parcel_id`	`VARCHAR(50)`	Cook County Assessor Property Index Number (PIN). Capture historical PIN variations where parcel splits/mergers occurred.
`address_current`	`VARCHAR(255)`	Modern standardized street address formatted to USPS standards.
`address_historic`	`JSONB`	Array of historical addresses with date ranges. **Mandatory Rule:** Must account for Chicago's 1909 and 1911 citywide street renumbering grids.
`latitude` / `longitude`	`DECIMAL(9,6)`	Fixed WGS-84 coordinate point representing the parcel centroid.
`footprint_geom`	`GEOMETRY(Polygon)`	PostGIS Polygon geometry in EPSG:3435 (State Plane Illinois East). Sourced from City GIS or Sanborn map digitisations.
`elevation_base`	`DECIMAL(6,2)`	Base ground elevation in feet above NAVD88. **Mandatory Rule:** Must track historical street grade raisings of 4–14 feet during the 1850s–1860s.
`neighborhood`	`VARCHAR(100)`	Current and historical sub-district designation (e.g., The Loop, River North, South Loop, Wolf Point).

B. Chronology & Temporal State Control

Attribute Name	Data Type	Specification & Data Collection Rules
version_id	UUID (PK)	Unique key representing a specific physical state/phase of a building on the property.
valid_from	DATE	Date this structural version became active/completed. Precision down to year mandatory; month
valid_to	DATE (Nullable)	Date this structural version was altered or demolished. NULL indicates the state is currently star
construction_start	DATE	Groundbreaking / permit issuance date for this specific construction phase.
construction_complete	DATE	Official completion or certificate of occupancy date.
demolition_date	DATE (Nullable)	Exact date demolition was completed or destruction occurred (e.g., October 9, 1871).
change_type	ENUM	Categorization of structural change: constructed, remodeled, reclad, expanded_vertic destroyed_fire, demolished.
predecessor_id / successor_id	UUID (FK)	Pointers linking structural versions on the same plot across time.

C. Physical & Architectural Specs

Attribute Name	Data Type	Specification & Data Collection Rules
name_current	VARCHAR(255)	Modern building name (e.g., Willis Tower, Acme Hotel).
name_historic	JSONB	Array of historical names with active date windows (e.g., Sears Tower [1973–2009], Berkshir
height_ft / stories	DECIMAL / INT	Architectural peak height in feet and number of above-ground floor levels.
floor_area_sqft / lot_area	DECIMAL(12,2)	Total gross interior floor space and lot surface area in square feet.
building_type	ENUM	Zoning/functional category: office, residential, hotel, retail, industrial, trans institutional, theater.
architectural_style	VARCHAR(100)	Ontology classification: Chicago School, Art Deco, Beaux-Arts, Prairie School, International Postmodern, Mid-Century Modern.
structural_system	VARCHAR(100)	Structural framing: Heavy Timber (Mill Construction), Load-Bearing Masonry, Cast/Wrought Steel Skeleton, Reinforced Concrete, Tube-in-Tube.
primary_materials	TEXT[]	Array of facade/exterior materials: Architectural Terra-Cotta, Indiana Limestone, Joliet Lemo Brick, Cast Iron, Glass Curtain Wall, Granite.
facade_color / cladding	VARCHAR(100)	Dominant color and finish details to guide procedural texture generation.
roof_type / features	VARCHAR(100)	Flat with parapet, Mansard, Hip, Gabled; roof features (water towers, cornices, clock towers,

D. People, Organizations & Legal Attributes

Attribute Name	Data Type	Specification & Data Collection Rules
architects	TEXT[]	Array of lead architects/firms (e.g., Burnham & Root, Adler & Sullivan, Holabird & Roche, Skidmore, Owings & Merrill, Ludwig Mies van der Rohe).
builders_contractors	TEXT[]	General contractors and structural engineers responsible for erection.

Attribute Name	Data Type	Specification & Data Collection Rules
`original_owner`	`VARCHAR(255)`	Entity commissioning initial construction.
`ownership_history`	`JSONB`	Deed transfer log: `[{owner: string, start_date: string, end_date: string, deed_ref: string}]`.
`notable_tenants`	`JSONB`	Anchor tenants and historical narrative figures: `[{tenant: string, start_year: int, end_year: int, story_notes: text}]`.
`historic_designation`	`JSONB`	Status records: Chicago Landmark (date), National Register of Historic Places (NRHP date), National Historic Landmark (NHL date).
`zoning_code`	`VARCHAR(50)`	Active zoning code during version lifespan (explains set-back rules, height limits).
`permit_numbers`	`TEXT[]`	City of Chicago Department of Buildings permit reference numbers.
`tax_assessment_history`	`JSONB`	Assessed valuations across time for investment tracking: `[{year: int, value_usd: numeric}]`.

E. 3D Spatial Assets & Visual Environment

Attribute Name	Data Type	Specification & Data Collection Rules
`3d_model_uri`	`VARCHAR(500)`	Path to optimization-ready spatial mesh (.glTF 2.0 / .bgltf / .fbx) representing this structural version.
`textures_uri`	`VARCHAR(500)`	Directory containing PBR texture maps (Albedo, Normal, Roughness, Metallic, Ambient Occlusion).
`lod_level`	`INT` (0-4)	Available Level of Detail: 0 (Extruded Footprint), 1 (Massing Model), 2 (Architectural Exterior Mesh), 3 (Detailed Ornamentation & Signage), 4 (Walkable Interior Atriums).
`roof_outline_geom`	`GEOMETRY(Polygon)`	PostGIS polygon for roof geometry, setbacks, and skylight penetrations.
`signage_history`	`JSONB`	Facade signage records: `[{text: string, type: string, installation_date: string, removal_date: string, mesh_uri: string}]`.
`street_level_assets`	`JSONB`	Associated era street furniture: period gas lamps/electric lights, hitching posts, sidewalk vaults, elevated "L" track structure linkages.

F. Archival Media & Data Quality Governance

Attribute Name	Data Type	Specification & Data Collection Rules
`photos`	`JSONB`	Archival image references: `[{url: string, date: string, photographer: string, library_source: string, rights: string}]`.
`drawings_blueprints`	`JSONB`	Original elevation drawings, floor plans, cross-sections, and structural detail blueprints.
`sanborn_maps`	`JSONB`	References to Sanborn Fire Insurance Maps: `[{volume: string, page: string, year: int, footprint_overlay_url: string}]`.
`confidence_score`	`INT` (1-100)	Data reliability metric. 100 = Modern terrestrial laser scan/Lidar; 80 = High-res photograph + Sanborn map + blueprints; 30 = Single 19th-century sketch/ drawing.

Attribute Name	Data Type	Specification & Data Collection Rules
`source_datasets`	`TEXT[]`	Provenance trail of all ingested primary datasets (e.g., UIC Special Collections, Chicago History Museum, HABS/HAER).
`last_verified`	`TIMESTAMP`	Date of last manual architectural validation by researcher.

3. Supporting Normalized Relational Tables

To prevent multi-value storage overhead and allow complex query filtering, researchers must populate the following normalized auxiliary tables linked to `property_id` or `version_id`:

Table Name	Primary Key / Foreign Keys	Table Purpose & Key Columns
`Renovation`	`renovation_id` (PK) `version_id` (FK)	Tracks non-demolition physical modifications. Columns: `start_date`, `completion_date`, `scope` (e. store-front modernization, elevator retrofit), `architect`, `cost_usd`.
`Tenant`	`tenant_id` (PK) `property_id` (FK)	Full occupancy tracking across centuries. Columns: `tenant_name`, `industry`, `start_date`, `end_da` `floors_occupied`, `cultural_note`.
`Media`	`media_id` (PK) `version_id` (FK)	Central media asset catalog. Columns: `media_type` (photo, blueprint, lithograph, video), `url`, `date_s` `archival_repository`, `license_status`.
`Event`	`event_id` (PK) `property_id` (FK)	Historical incidents attached to space. Columns: `event_date`, `event_title`, `description` (e.g., Eastland disaster response, Great Fire line, Eastland pier), `source_url`.
`OwnershipRecord`	`record_id` (PK) `property_id` (FK)	Chain of title deeds. Columns: `grantor`, `grantee`, `sale_date`, `sale_price_usd`, `book_page_re`
`UserContribution`	`contrib_id` (PK) `property_id` (FK)	Crowd-sourced archival edits. Columns: `user_id`, `field_edited`, `old_value`, `new_value`, `submission_date`, `verification_status`.

4. Primary Data Ingestion Sources for Researchers

Geographic & Physical Records

- **Chicago Data Portal:** Modern building footprints, 3D building heights, current PIN zoning parcel GIS layers.
- **Cook County Assessor & Recorder of Deeds:** Deed chains, PIN historical histories, property tax bounds.
- **Sanborn Fire Insurance Maps (Library of Congress):** Crucial 1886–1950 volume maps giving precise footprints, wall thickness, construction materials (brick vs. frame), and elevator shafts.
- **USGS 3DEP / Illinois Height Modernization:** Airborne LiDAR point clouds for modern high-precision envelope geometry.

Architectural & Photographic Archives

- **Chicago History Museum (CHM):** Hedrich-Blessing architectural photography collection, pre-fire lithographs.
- **Art Institute of Chicago (Burnham & Ryerson Libraries):** Original architectural drawings from Burnham, Sullivan, Roche, and Jenney.
- **UIC Special Collections & CPL Digital:** Historic postcard collections, neighborhood urban renewal photographic surveys.
- **ProQuest Historical Chicago Tribune:** Construction announcements, building permit notices, historic retail advertising.

5. Implementation Stack & Architectural Pipeline

- **Database Layer:** PostgreSQL 16+ with `PostGIS` extension for spatial queries and `temporal_tables` extension for automated system versioning.
- **ETL Data Pipeline:** Python 3.11+ using `GeoPandas`, `Shapely`, and `SQLAlchemy` to ingest GIS Shapefiles, process GeoJSON footprints, and normalize tax maps.
- **Data API Service:** GraphQL endpoint accepting bounding-box coordinates and target date parameters (e.g., `query { buildings(bbox: [...], year: 1893) { id name spatialMeshUri } }`).
- **3D Engine Renderer:** WebGL via **CesiumJS** (optimized for geo-referenced scale, global terrain, and time-scrubbing sliders), **Three.js** (for browser walk-throughs), or **Unreal Engine 5** (for ultra-high-fidelity offline VR/desktop walkthroughs).

6. Mandatory Chicago-Specific Research Directives

CRITICAL HISTORICAL RULES RESEARCHERS MUST ENFORCE

1. **The 1871 Great Chicago Fire (October 8–10, 1871):** The fire wiped out nearly all downtown structures between the Chicago River and Lake Michigan up to Fullerton. Treat October 1871 as an absolute hard temporal reset. All building versions in this zone must terminate on 1871-10-09, with new versions commencing during the reconstruction boom (1872–1878).
2. **The 1909 / 1911 Street Renumbering:** Edward P. Brennan's renumbering plan altered virtually every street address in Chicago in 1909 (and the Loop core in 1911). Researchers *must never* use pre-1909 addresses directly for GIS placement. Every pre-1909 address must be cross-referenced through the 1909 Plan of Chicago renumbering index tables to map to modern coordinates.
3. **Street Grade Elevation Raising (1850s–1860s):** Downtown Chicago raised its street levels by 4 to 14 feet to install sewers and solve drainage issues. Buildings constructed prior to 1855 had their ground floors converted into basements, and new entrances constructed at the new grade. Base elevation data must reflect the active year.
4. **Infrastructure Layers (The "L" and Freight Tunnels):** Downtown walkthroughs must incorporate contextual transit: Union Loop Elevated tracks erected 1895–1897; Chicago Tunnel Company 60-mile narrow-gauge underground freight network constructed beneath Loop streets starting 1899.